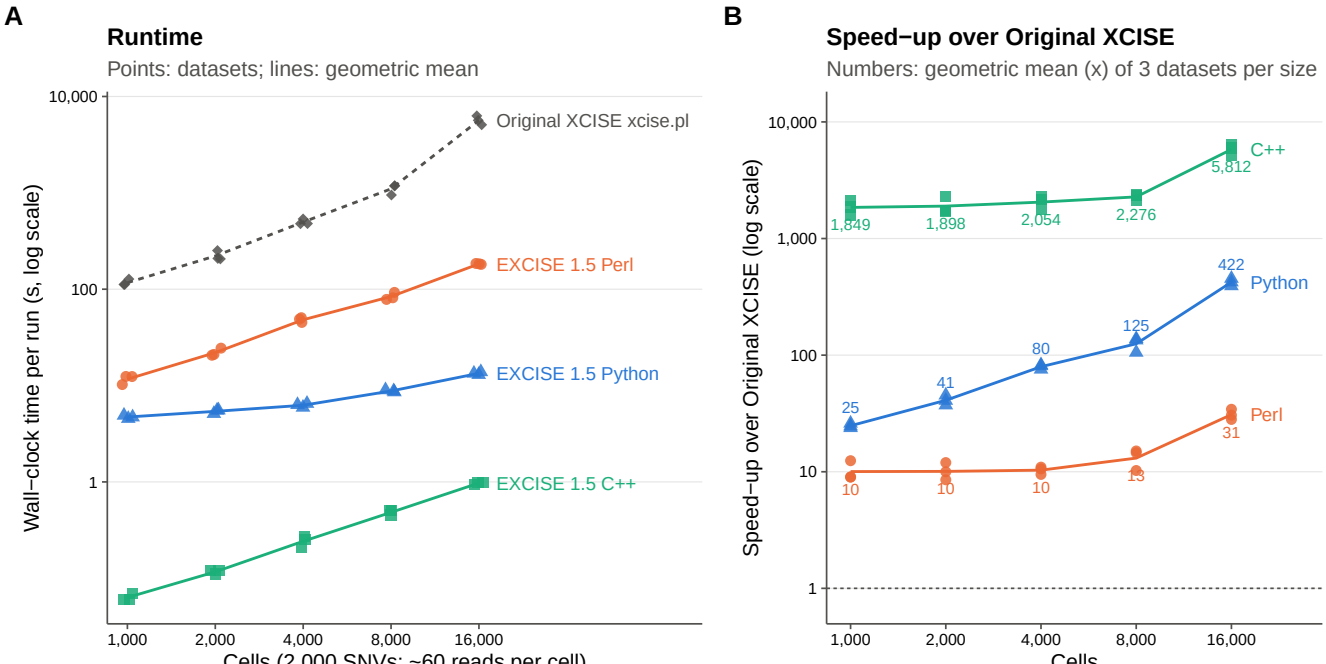

All tools: 4 tries (−t 4) on the same simulated BAMs (15 datasets), run one at a time; times are full runs including BAM reading.
EXCISE 1.5: −j 4 −pretrain. Original XCISE (Vityay/XCISE @3a1d063): −u 4 (same SNV filter); no −j or −pretrain, tries run one after another.
Python includes one−off numba compilation. Optimiser alone: Python 0.04−0.06 s, C++ at most 0.01 s.
Cell accuracy vs simulated truth (mean): Original 0.998, Perl 0.998, Python 0.998, C++ 0.998.
bc2xci identical to Perl EXCISE 1.5: Python 10/15, C++ 12/15 runs; the rest reached different local optima (at most 1 cell labelled differently).